CHAPTER 8

Square-Root Coordinates and Spherical Geometry

The categorical Fisher metric has a concrete global model. Taking square roots of the probabilities sends the positive simplex to an orthant of a round sphere. The metric becomes ordinary spherical length, making exact distances and shortest paths accessible through Euclidean vector calculations.

Imagine drawing a straight segment between two points of the probability triangle. That segment is a natural mixture of the endpoint laws, but it need not be the shortest path for the Fisher metric. Square-root coordinates reveal why: the metric measures length on a curved spherical surface. The distinction between a straight mixture and a shortest geometric path will be visible in both drawings.

8.1 The square-root isometry

For a positive categorical law define

$$s_i = 2\sqrt{p_i}. \tag{8.1}$$

Normalization gives $\sum_i s_i^2 = 4$, and positivity gives $s_i > 0$. Thus s belongs to the positive orthant of the radius-2 sphere in $\mathbb{R}^n$.

Differentiate the map:

$$ds_i = \frac{dp_i}{\sqrt{p_i}}, \qquad \sum_i ds_i^2 = \sum_i \frac{dp_i^2}{p_i} = ds_F^2.$$

The Euclidean inner product on the image tangent spaces is exactly the Fisher inner product. The map is one-to-one, with inverse $p_i = s_i^2/4$.

Proposition 8.1 (Square-root representation)**.** *The positive categorical simplex with the Fisher metric is isometric to the positive orthant of the round sphere $S^{n-1}(2)$.*

Proof. The map and its inverse are smooth on their positive domains, and the calculation above proves preservation of tangent inner products. For a tangent displacement, $\sum_i dp_i = 0$ also gives $s \cdot ds = 2\sum_i dp_i = 0$, confirming that the image displacement is tangent to the sphere. □

An *isometry* preserves the lengths and angles measured by the two metrics. It also preserves lengths of curves, since those lengths are integrals of tangent norms. The flat triangle remains useful as a probability diagram; the sphere displays the geometry supplied by Fisher information.

Example 8.1 (A distribution and a tangent vector on the sphere)**.** Map $P = (1/2, 1/3, 1/6)$ to the sphere. Describe the image of a tangent vector v.

Solution. The image is $(\sqrt{2}, 2/\sqrt{3}, 2/\sqrt{6})$, whose squared norm is $2 + 4/3 + 2/3 = 4$. The image velocity has components $v_i/\sqrt{p_i}$. Its dot product with the image point is $2\sum_i v_i = 0$, so it lies in the tangent plane of the sphere.

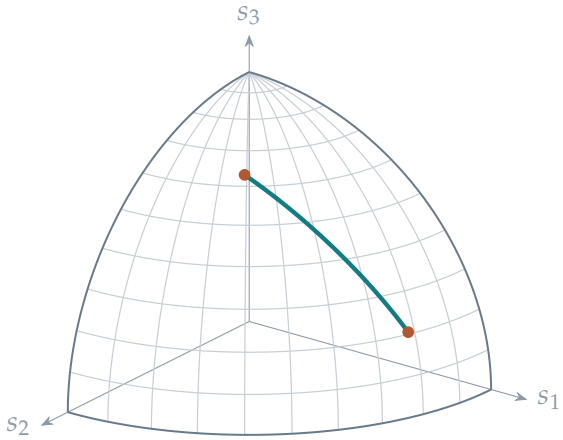


Figure 8.1. An orthographic view of the positive octant of the radius-2 sphere. The teal curve is the great-circle image of a categorical Fisher geodesic; orange points mark its endpoint laws.

8.2 Why shortest spherical paths are great-circle arcs

A great circle is the intersection of a sphere with a plane through its center. Two non-antipodal points determine a shorter great-circle arc, with length equal to radius times central angle.

The minimizing property follows from a useful length estimate. Let $c(t)$ lie on a radius-R sphere, let e_0 be the unit vector toward its starting point, and define the central angle $\theta(t) = \arccos(e_0 \cdot c(t)/R)$. At regular interior angles,

$$|\theta'| = \frac{|e_0 \cdot c'|}{R \sin\theta} \le \frac{\|c'\|}{R}.$$

For the inequality, project e_0 into the tangent plane at $c(t)$; that projection has norm $\sin\theta$, while c' is tangent. Cauchy–Schwarz then gives the bound. Integrating, and

taking endpoint limits when necessary, shows that every connecting curve has length at least R times the endpoint central angle. The shorter great-circle arc attains this lower bound.

For square-root images of categorical laws, the endpoint angle is at most $\pi/2$. Their shorter arc has nonnegative linear-combination coefficients, as the explicit formula below shows. It therefore remains in the spherical orthant. The spherical shortest path is also an admissible path in the statistical family.

8.3 The Bhattacharyya coefficient and Fisher distance

On the sphere, the shortest distance between two points is determined by the angle between their radius vectors. The square-root map lets us calculate that angle directly from the original probabilities. Its cosine is a familiar statistical overlap: the sum of the geometric means of corresponding category weights. For laws $P = (p_i)$ and $R = (r_i)$, define their *Bhattacharyya coefficient*

$$B(P, R) = \sum_i \sqrt{p_i r_i}.$$

Cauchy–Schwarz applied to the unit vectors $\sqrt{P}$ and $\sqrt{R}$ gives $B \leq 1$, with equality exactly when $P = R$. Nonnegativity gives $B \geq 0$. At the boundary $B = 0$ exactly when their supports are disjoint.

The square-root image vectors $s = 2\sqrt{P}$ and $r = 2\sqrt{R}$ satisfy

$$\cos \alpha = \frac{s \cdot r}{4} = B(P, R).$$

Multiplying the central angle by the radius gives

$$\boxed{d_F(P, R) = 2 \arccos B(P, R).} \tag{8.2}$$

The formula is symmetric, vanishes precisely at equality, and satisfies the triangle inequality because it is a shortest-path distance.

Example 8.2 (Computing a categorical distance)**.** For $P = (0.5, 0.3, 0.2)$ and $R = (0.2, 0.5, 0.3)$, express the coefficient and distance.

Solution. The coefficient is

$$B = \sqrt{0.10} + \sqrt{0.15} + \sqrt{0.06}.$$

Numerically $B \approx 0.948475$, so $\alpha \approx 0.322408$ radians and $d_F \approx 0.644817$. Each term compares the mass assigned to the same category in the two laws; the sum is their square-root-vector overlap.

8.4 Constructing the constant-speed geodesic

Let $P \neq R$ be positive laws and $\alpha = \arccos B(P, R)$. Their spherical interpolation is

$$s(t) = \frac{\sin((1-t)\alpha)}{\sin \alpha} s(0) + \frac{\sin(t\alpha)}{\sin \alpha} s(1), \qquad 0 \leq t \leq 1, \tag{8.3}$$

where $s(0) = 2\sqrt{P}$ and $s(1) = 2\sqrt{R}$. The probability path is

$$\gamma_i(t) = \frac{s_i(t)^2}{4}. \tag{8.4}$$

It is a *geodesic*: a shortest path in the Fisher geometry, parameterized with constant speed.

To verify its properties, choose orthonormal vectors

$$e_0 = \frac{s(0)}{2}, \qquad e_1 = \frac{s(1)/2 - \cos \alpha \, e_0}{\sin \alpha}.$$

The interpolation becomes

$$s(t) = 2\big[\cos(t\alpha) e_0 + \sin(t\alpha) e_1\big].$$

Thus its squared norm is 4, its speed is 2α, and its length over $[0, 1]$ is 2α. The sine coefficients in (8.3) are positive between the endpoints, so every coordinate stays positive. If $P = R$, the appropriate geodesic is the constant path.

Example 8.3 (The Fisher midpoint)**.** Find the probability midpoint $\gamma(1/2)$ between two positive laws.

Solution. At the midpoint both coefficients in (8.3) equal $1/[2\cos(\alpha/2)]$. Consequently

$$\gamma_i(1/2) = \frac{(\sqrt{p_i} + \sqrt{r_i})^2}{2(1 + B(P, R))}. \tag{8.5}$$

The numerators sum to $2 + 2B$, confirming normalization. This midpoint generally differs from the ordinary average $(P + R)/2$.

8.5 Mixture paths and geometric paths

The mixture $M(t) = (1-t)P + tR$ traces a straight segment in probability coordinates. Its Fisher length is

$$L_F(M) = \int_0^1 \sqrt{\sum_i \frac{(r_i - p_i)^2}{(1-t)p_i + t r_i}} \, \mathrm{d}t. \tag{8.6}$$

The shortest-path property implies $L_F(M) \geq d_F(P, R)$. Equality holds when the mixture traces the same minimizing arc, with no backtracking. For general three-outcome

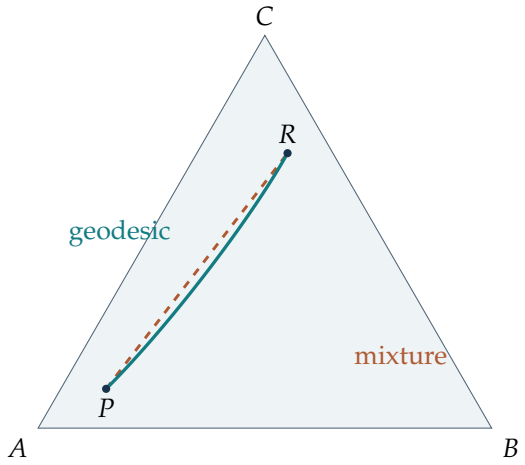


Figure 8.2. The straight mixture and Fisher geodesic from $P = (.8, .1, .1)$ to $R = (.1, .2, .7)$ in barycentric coordinates. Their square-root images trace different spherical routes.

endpoints its spherical image bends away from the great circle, and the inequality is strict.

A useful special case fixes the conditional proportions of a group of categories. For example, let

$$P = (a, (1-a)r, (1-a)(1-r)),$$

and let R have the same form with a replaced by b, keeping r fixed. Their square-root vectors lie in the plane spanned by $(1, 0, 0)$ and $(0, \sqrt{r}, \sqrt{1-r})$. Varying only the first probability traces a great-circle arc in that plane. A monotone mixture therefore has the geodesic's image, although its speed in spherical angle is generally variable.

Path image and parameterization are separate features. Two parameterizations can trace the same shortest route at different speeds. A constant-speed geodesic fixes both the route and its speed over the chosen parameter interval.

8.6 Bernoulli angles and arc coordinates

For $P_p = (p, 1-p)$, the image is

$$s(p) = (2\sqrt{p}, 2\sqrt{1-p}).$$

Writing $p = \sin^2 \varphi$ gives $s = (2\sin\varphi, 2\cos\varphi)$, where $0 < \varphi < \pi/2$. The ordinary central angle measured from the second coordinate axis is

$$\varphi = \arcsin \sqrt{p}.$$

Since the radius is 2, the arc coordinate is $\theta = 2\varphi = 2\arcsin\sqrt{p}$. This is the length coordinate of Chapter 7. The central angle and arc length differ by the radius factor.

Example 8.4 (A Bernoulli midpoint)**.** Find the constant-speed midpoint between $p = 1/4$ and $p = 3/4$.

Solution. The central angles are $\pi/6$ and $\pi/3$. Their midpoint is $\pi/4$, so the midpoint probability is $\sin^2(\pi/4) = 1/2$. The total arc length is $2(\pi/3 - \pi/6) = \pi/3$.

Every monotone Bernoulli probability path has the same quarter-circle route between its endpoints. This one-dimensional case explains why a linear probability mixture can have minimizing length while failing to have constant Fisher speed.

8.7 Chord length and Hellinger distance

The straight Euclidean chord between square-root vectors has length

$$\|2\sqrt{P} - 2\sqrt{R}\|^2 = 4\sum_i (\sqrt{p_i} - \sqrt{r_i})^2 = 8(1 - B). \tag{8.7}$$

The *Hellinger distance*, in a common normalization, is

$$h(P, R) = \frac{1}{\sqrt{2}}\|\sqrt{P} - \sqrt{R}\|, \qquad h(P, R)^2 = 1 - B(P, R).$$

It is a metric because it is a scaled Euclidean distance under an injective map. The chord is $2\sqrt{2}\,h$, while the intrinsic Fisher distance is $2\arccos(1 - h^2)$.

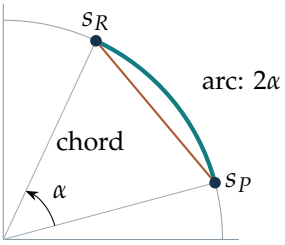


Figure 8.3. Arc and chord lengths on the radius-2 circle. The arc stays within the statistical manifold; the interior of the chord lies inside the sphere.

For small central angle, the chord $4\sin(\alpha/2)$ and arc 2α agree to first order. Expanding square roots gives $h(P, P + v)^2 = \frac{1}{8}\sum_i v_i^2/p_i + O(\|v\|^3)$, consistent with the same local Fisher geometry after accounting for this normalization.

8.8 Initial velocity and local consistency

The explicit geodesic formula also specifies the initial direction in probability coordinates. Differentiate (8.3) at $t = 0$, then use $\gamma_i' = s_i s_i'/2$. For distinct endpoints,

$$\gamma_i'(0) = \frac{2\alpha}{\sin\alpha}\left(\sqrt{p_i r_i} - B(P, R)p_i\right). \tag{8.8}$$

The components sum to zero because both summed terms equal B. The direction is a tangent vector based at P, and its Fisher norm is 2α. Thus an endpoint pair determines both a shortest route and the exact initial velocity for traversing it over a unit interval.

For nearby laws, the global formula recovers the local Fisher metric. Set $R = P + h$ with $\sum_i h_i = 0$. Expanding the square root in each overlap term gives

$$\begin{aligned} B(P, P+h) &= \sum_i p_i \sqrt{1 + h_i/p_i} \\ &= 1 - \frac{1}{8} \sum_i \frac{h_i^2}{p_i} + O(\|h\|^3). \end{aligned}$$

Since $\cos\alpha = 1 - \alpha^2/2 + O(\alpha^4)$, the squared distance satisfies

$$d_F(P, P+h)^2 = \sum_i \frac{h_i^2}{p_i} + O(\|h\|^3).$$

The arc, chord, and local metric calculations therefore agree to their appropriate leading orders. Their global differences arise as the displacement grows.

Example 8.5 (A geodesic with a persistent category symmetry)**.** Connect $P = (1/3, 1/3, 1/3)$ to $R = (1/2, 1/4, 1/4)$ with constant Fisher speed.

Solution. Both endpoints have equal second and third coordinates, so their square-root vectors lie in the plane $s_2 = s_3$. The connecting great circle remains in that plane. Every intermediate law has the form $(a, (1-a)/2, (1-a)/2)$, a Bernoulli split between A and the combined BC group.

Let $\varphi_0 = \arcsin\sqrt{1/3}$ and $\varphi_1 = \arcsin\sqrt{1/2}$. The constant-speed path is

$$a(t) = \sin^2\big((1-t)\varphi_0 + t\varphi_1\big), \qquad \gamma(t) = \left(a(t), \frac{1-a(t)}{2}, \frac{1-a(t)}{2}\right).$$

Its total length is $2(\varphi_1 - \varphi_0)$. Grouping the two symmetric categories reduces this calculation to the one-dimensional circular geometry without changing the Fisher length.

The probability coordinates of a constant-speed geodesic generally change at non-constant rates. It is the angle on the sphere, rather than each probability separately, that advances uniformly. The construction specifies a probability model at each intermediate position and therefore permits the numerical readings, entropy, or likelihood of that model to be evaluated along the path.

8.9 Boundary completion and curvature

The positive spherical orthant has boundary points at finite distance. Adding its nonnegative boundary produces the *metric completion*: every Cauchy sequence then has

a limit. The square-root inverse identifies these added points with laws having zero probabilities.

Example 8.6 (Two pure distributions)**.** Take $P = (1,0,0)$ and $R = (0,1,0)$ in the completion.

Solution. Their square-root vectors $(2,0,0)$ and $(0,2,0)$ have central angle $\pi/2$. Hence $d_F(P,R) = \pi$. Their ordinary probability-vector distance is $\sqrt{2}$, their square-root chord length is $2\sqrt{2}$, and both directed KL divergences are infinite. The Fisher distance remains finite because a finite-length spherical arc connects the limiting states.

For three outcomes the image is an open patch of a 2-sphere of radius 2. A round radius-R sphere has Gaussian curvature $1/R^2$, so the categorical Fisher surface has constant curvature $1/4$. For $n \geq 3$, the corresponding higher-dimensional sphere has the same sectional curvature in every two-dimensional tangent plane. A Bernoulli family is one-dimensional; it has arc length geometry, and Gaussian curvature is not defined for it.

Shortest paths answer a geometric question about prescribed endpoints. The next chapter uses the same metric to define motion driven by a scalar objective.

8.10 Exercises

8.1 Prove the square-root map preserves tangent inner products and maps the normalization tangent space to the spherical tangent space.

8.2 Map $P = (1/4, 1/4, 1/2)$ to $S^2(2)$ and verify the norm directly.

8.3 For $P = (0.5, 0.3, 0.2)$ and $R = (0.2, 0.5, 0.3)$, compute B, the central angle, Fisher distance, and chord length numerically.

8.4 Prove $0 \leq B(P,R) \leq 1$ and characterize equality at both ends.

8.5 Derive the geodesic midpoint formula (8.5). Verify its coordinates sum to one.

8.6 Use the orthonormal representation to prove that (8.3) has constant speed 2α.

8.7 For Bernoulli probabilities $1/4$ and $3/4$, compute the central-angle separation, Fisher distance, and square-root chord length.

8.8 Explain why the shorter spherical arc stays in the positive orthant when both endpoints are positive.

8.9 Derive the mixture-length formula (8.6). Explain why its value is at least the Fisher distance.

8.10 Verify that the fixed-conditional-proportion family in the text lies on a great circle. Explain why its monotone mixture paths have minimizing length.

8.11 Compute the geodesic from $(1,0,0)$ to $(0,1,0)$ in the metric completion, including its probability parameterization.

8.12 Derive $h^2 = 1 - B$, the chord formula, and $d_F = 2\arccos(1-h^2)$.

8.13 Show that the distance between any two categorical laws in the completion is at most π. Characterize pairs attaining this bound.

8.14 Explain why divergence of a coordinate metric coefficient does not by itself imply infinite boundary distance. Use the Bernoulli integral as evidence.

8.15 *Exploration.* Compare the mixture and Fisher paths from $(0.8, 0.1, 0.1)$ to $(0.1, 0.2, 0.7)$. Numerically integrate the mixture length and compare midpoints.

8.16 Expand $(\sqrt{p_i + v_i} - \sqrt{p_i})^2$ through quadratic order and obtain the local Hellinger approximation.

8.17 Prove directly from the spherical distance formula that simultaneous permutation of the outcome labels preserves Fisher distance.

8.18 *Exploration.* Move two distributions toward different vertices. Track B, chord length, Fisher distance, and both KL divergences. Identify which limits depend on support.

Selected Solutions

Chapter 8

8.1 The differential sends v_i to $v_i/\sqrt{p_i}$, giving the Fisher inner product by an ordinary dot product. Its dot product with $2\sqrt{P}$ is $2\sum_i v_i = 0$.

8.3 The coefficient is $B \approx 0.948475$. The central angle is 0.322408 radians, Fisher distance is 0.644817, and chord length is 0.642028. Evaluate the formulas before rounding to keep their small differences visible.

8.5 The half-angle identity gives $\sin(\alpha/2)/\sin\alpha = 1/[2\cos(\alpha/2)]$. Squaring the midpoint square-root coordinates gives (8.5), with total numerator $2 + 2B$.

8.7 The separation is $\pi/6$, the Fisher distance is $\pi/3$, and the chord is $4\sin(\pi/12) = \sqrt{6} - \sqrt{2}$.

8.9 Along the mixture the derivative is $R - P$. Substitute it into the Fisher speed and integrate. The geodesic distance is the infimum of admissible path lengths, giving the inequality.

8.11 The square-root geodesic is $(2\cos(\pi t/2), 2\sin(\pi t/2), 0)$. The probability path is $(\cos^2(\pi t/2), \sin^2(\pi t/2), 0)$, of constant Fisher speed π.

8.13 The coefficient B lies in $[0, 1]$, so $2\arccos B \le \pi$. Equality holds precisely for disjoint supports.

8.17 A simultaneous permutation merely reorders the summands $\sqrt{p_i r_i}$, leaving their sum and the distance unchanged.